

Azul Checkpoint #2



Group 5



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Checkpoint #1 Feedback

Feedback

Preliminary analysis informed our investigation and developed our understanding of the project.

Our presentation focused on the relationship between different aircraft and their fuel burn during different routes or TOW.

Problem 1

Fuel efficiency was not well characterized during our first presentation, lacking a clear explanation of what our investigation meant.

Problem 2

The proposed adjustments were overly generalized and were not actionable items. Since then, our continued analysis has helped us draw more specific conclusions.

Clear analysis

Previously our conclusions were too broad, due to the surface level nature of the preliminary analysis. With more specific insights from the indepth analysis, we will lay out clear actionable items, such as switching which aircraft fly certain routes.

Actionable Items

To make our analysis more clear, we define the scope and intent of each included graph. The structured analysis we plan to continue with will form a fluid cohesive argument for how Azul can optimize their network.



Main research question

How can Azul identify implementable aircraft reassignment strategies across route groups with overlapping operational sweetspots in order to improve operational efficiency without assuming full fleet replacement?



Subquestions

1. Which city pairs or route groups exhibit overlapping operational sweetspots between aircraft types, making reassignment operationally plausible?
2. To what extent do demand fluctuations across routes or seasons justify the reassignment of aircraft types within these overlapping sweetspots?
3. How do airport infrastructure and operational constraints limit the feasibility of using larger or alternative aircraft types on specific routes?
4. How much potential benefit could these reassignment strategies generate for the airline, based on the operational metrics available in the dataset?
5. How sensitive are the identified reassignment opportunities to the analytical assumptions used, particularly seat-based swapability, sweetspot thresholds, and sample size?



Questions based on methodology, validation, or limitations

- Whether payload composition may explain part of the observed differences between aircraft types.
- Whether the selected variables are appropriate for comparing operational fit.
- Whether the analysis contains blind spots or biases.
- Whether seat count is an adequate proxy for swapability.
- Whether the joint sweetspot sample is large enough to support meaningful conclusions.

Project roadmap and work division

Step 01

Data foundation
Burak + Eduardo
Cleaning,
assumptions,
and comparable
TOW bands

Step 02

Checkpoint #1
refinement
Robert + Alexandre
Clearer visuals and
more comparable
evidence

Step 03

Deep-dive analyses
Burak + Robert
Sweetspots, route
groups,
and seasonal trade
studies
.

Step 04

Optimization
modeling
Eduardo + Alexandre
Theoretical fuel
model,
capacity screen,
savings

Goal

All four members
Actionable
reassignment
options and final
story

RACI matrix project management

Legend

R =
Responsible

A =
Accountable

C =
Consulted

I =
Informed

	Burak	Robert	Alexandre	Eduardo
Data foundation and cleaning	R	A	I	C
Checkpoint #1 refinement and visualization updates	C	I	A	R
Deep-dive analyses: sweetspots, route groups, seasonality	A	A	C	R
Optimization modeling: theoretical fuel model, capacity screen, savings	C	R	A	I
Final synthesis: recommendations and presentation story	R	C	C	A

1. Data filtering and validation

- Filtering count chart and before-vs-after scatterplots to show how cleaning changes the dataset and improves reliability.

2. Within-aircraft operating relations

- Scatterplots with trendlines for TOW vs fuel burn, distance vs trip time, and trip time vs fuel burn to identify operating profiles and efficiency deterioration points.

3. Comparable aircraft-to-aircraft analysis

- Overlay scatterplots, sweetspot range charts, and capacity-screening tables to compare aircraft only in overlapping TOW, distance, and route conditions.

4. Route and seasonal trade studies

- Route heatmaps, monthly trend charts, and route-group comparison tables to test where reassignment is operationally realistic.

Any questions?



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Ask away!

Thanks for the attention!!